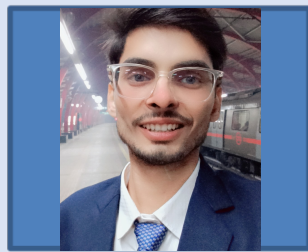




DELHI TECHNICAL CAMPUS

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ARCHITECTURE AND DEVELOPMENT OF AN ENTERPRISE-GRADE AI CHATBOT WITH SEMANTIC KNOWLEDGE RETRIEVAL

Introduction

The Enterprise AI Chatbot is a modern, privacy-centric solution engineered to address the critical challenge of information discovery within vast, unstructured PDF repositories. In contemporary environments, organizations generate massive amounts of "Dark Data" critical information trapped in technical manuals and research logs that is notoriously difficult to navigate. Traditional search tools, relying exclusively on keyword matching, frequently fail in these complex contexts as they lack the semantic understanding required to interpret a user's underlying intent. To bridge this gap, this project implements a sophisticated Retrieval-Augmented Generation (RAG) architecture. This system moves beyond simple pattern matching by utilizing natural language processing to convert document text into high-dimensional vector embeddings, allowing the AI to "understand" the conceptual relationship between a question and the stored content. When a query is made, the system performs a semantic similarity search to retrieve the most relevant document chunks, which are then processed by a Large Language Model (LLM) to generate a precise, fact-grounded answer. A core pillar of this project is absolute data sovereignty; by hosting the Llama 3.2 model locally via the Ollama framework, the system ensures that sensitive intellectual property never leaves the internal network. Integrating ChromaDB for vector storage and a FastAPI-driven backend creates a robust "Corporate Brain" that transforms static archives into a dynamic, secure, and interactive knowledge base.

Objectives

The primary objectives of this project center on architecting a secure, high-performance AI ecosystem capable of transforming static document archives into an interactive knowledge asset. A fundamental goal is the development of a locally-hosted AI assistant that utilizes Retrieval-Augmented Generation (RAG) to perform deep semantic knowledge retrieval from unstructured PDF repositories. This involves moving beyond rudimentary keyword matching to a system that understands the conceptual intent of a query, ensuring that every generated response is contextually aware and strictly verifiable against the source material. To guarantee absolute data privacy and sovereignty, the project is designed to operate entirely within an organization's internal network, utilizing local Large Language Model (LLM) inference to prevent sensitive intellectual property from being exposed to external cloud providers. Furthermore, the objective includes a strong focus on user experience, aiming to design a modern, intuitive web interface rooted in Glassmorphism principles that bridges the gap between complex backend engineering and professional accessibility. Ultimately, the project seeks to establish a dual-mode processing framework that offers seamless transitions between document-specific Q&A and general intelligence, providing a scalable and deployment-ready solution that enhances organizational efficiency while maintaining rigorous security standards.

Feasibility Study and Tool Used

Feasibility Study:

The technical feasibility of this project is established through the successful integration of Llama 3.2 with ChromaDB using a Python-based orchestration layer. By utilizing the Ollama framework, the system proves that enterprise-grade AI performance is achievable on local hardware without relying on external cloud APIs. Operationally, the project replaces manual searching with an automated RAG pipeline, featuring a "one-click" deployment script for easy maintenance by non-technical staff. Economically, the solution is highly viable; leveraging an open-source stack—including FastAPI and LangChain—eliminates recurring subscription fees and mitigates the massive financial risks associated with third-party data breaches.

Tools and Technologies Used:

- Core Architecture: Retrieval-Augmented Generation (RAG)
- AI Engine: Llama 3.2 (via Ollama local inference)
- Backend: FastAPI (Python 3.8+)
- Vector Database: ChromaDB (for 768-dimension embeddings)
- NLP Tools: LangChain and pypdf (for extraction and chunking)
- Frontend UI: HTML5, CSS3, JavaScript (Glassmorphism design)

System Architecture

1. Frontend Layer

The frontend is built using a modern stack of HTML5, CSS3, and JavaScript, featuring a responsive web interface designed with Glassmorphism principles for high visual clarity. It facilitates a seamless user experience by providing a dual-mode interaction area where users can toggle between general AI chat and document-specific querying.

2. Backend Service

The backend is powered by the FastAPI framework, managing high-performance asynchronous request handling and orchestrating the entire RAG pipeline from ingestion to response generation. It serves as the central hub for local LLM inference via Ollama, ensuring all data processing remains securely within the internal network. This layer utilizes LangChain for recursive character splitting and automated text cleaning to ensure high data quality for the AI model. Finally, it integrates dual-mode routing logic to switch between document-specific and general AI responses, providing a seamless bridge between raw data and the intelligent user interface.

3. Database Design:

The system utilizes ChromaDB, an AI-native vector database, to store high-dimensional embeddings for sub-second semantic similarity searches. This specialized design allows the chatbot to instantly retrieve relevant technical context from thousands of stored segments, providing a scalable foundation for organizational knowledge. By maintaining a local vector index, the database ensures lightning-fast, offline retrieval while keeping sensitive data securely within the internal network.

Design and Implementation

Design Methodology

The system follows a modular RAG architecture that separates document ingestion from query processing to ensure a scalable and secure workflow. The interface employs Glassmorphism principles, using semi-transparent elements to provide a modern, professional, and intuitive user experience.

Document Processing & Ingestion

The pipeline uses the pypdf library for text extraction and a recursive character splitter to break content into overlapping 1,000-character chunks. This specific implementation preserves technical context across segments, ensuring that vital information is not lost during the data preparation phase.

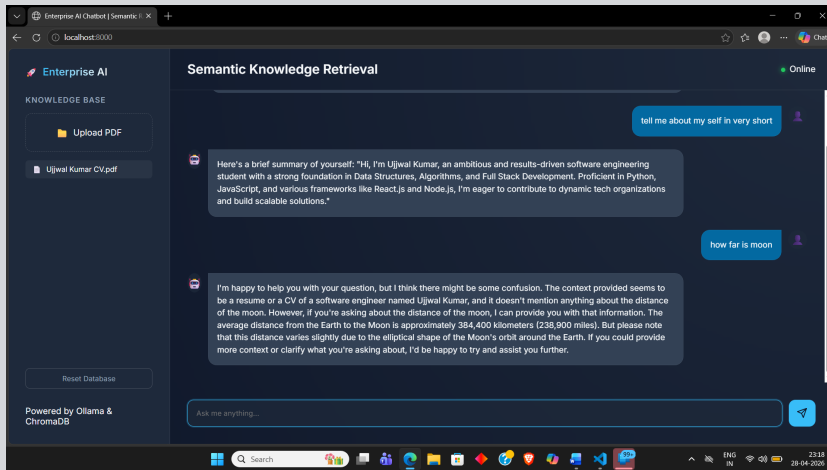
Vector Storage & Embedding

ChromaDB is utilized to store high-dimensional embeddings, allowing for sub-second semantic similarity searches using Cosine Similarity metrics. This enables the system to mathematically identify and retrieve the most relevant sections of a document to ground the AI's responses.

Core Logic & AI Integration

The backend uses FastAPI and LangChain to manage a dual-mode logic that switches between document-grounded answers and general AI chat. By hosting the Llama 3.2 model via Ollama, the implementation ensures all inference remains entirely offline and securely within the internal network.

Result



Conclusion

In conclusion, the RAG-based chatbot architecture successfully improves information retrieval speed and accuracy compared to traditional search methods. The integration of local LLMs ensures absolute data sovereignty while providing an intuitive, human-like interface for interacting with complex data. This project demonstrates how modern AI can be deployed ethically and efficiently within a private enterprise environment.

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